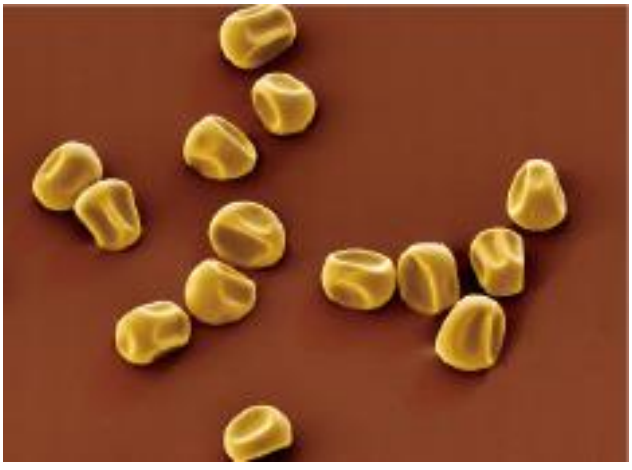
Structured questions

|!|BCELB0130000001|!| ★

The photomicrographs below show the pollen grains of plant A and plant B.



Pollen grains of plant A (×180)



Pollen grains of plant B (×180)

(a) With reference to the photomicrographs, deduce the agents of pollination of plant A and plant B. (4 marks)

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(b) Plant B can be separated into male plant and female plant, in which each type of plant only has either male or female reproductive structures.

(i) Explain the advantage of having separate male and female plants. (3 marks)

(ii) Suggest ***one*** disadvantage of having separate male and female plants. (1 mark)

##

(a) Plant A is pollinated by wind. (1)
The pollen grains are smooth / smaller in size and therefore can be carried by wind easily. (1)
Plant B is pollinated by insects. (1)
The pollen grains are rough and therefore can be attached on insects easily. (1)

(b) (i) This ensures cross-pollination / avoids self-pollination. (1)
The genetic variation of the offspring is greater, (1)
which allows a better adaptation to environmental change. (1)

(ii) Any ***one*** of the following: (1)
- The chance of pollination is low if the male and female plants are separated far way. (1)
- Only female plants can produce offspring / fewer plants can produce offspring. (1)
- External agents are needed to carry pollen grains from male plants to female plants. (1)

(Or other reasonable answers)

(8 marks)

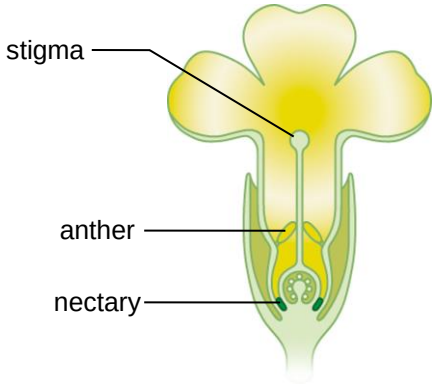
##

	Skill assessed	Mark(s)
(a)	Analysing data	4
(b)(i)	Applying concepts	3
(b)(ii)	Applying concepts	1

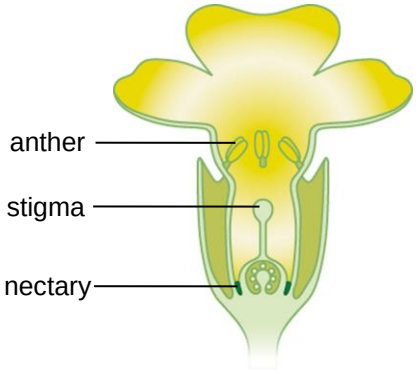
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|!|BCELB0130000002|!| ★

A primrose plant produces two types of flowers. Type X flowers have high stigmas and low anthers, and type Y flowers have low stigmas and high anthers.



Type X flower



Type Y flower

(a) State ***two*** observable features of type X flower, other than nectaries, showing that it is an insect-pollinated flower. (2 marks)

(b) Butterflies use their long mouthparts to collect nectar from the primrose flower. Describe and explain how they help cross-pollinate the flowers when they visit between these types of flowers.

(3 marks)

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(c) What is the advantage of having the anthers and stigmas at different positions in the primrose flowers? (2 marks)

##

(a) Brightly coloured / large petals (1)
Anthers and stigma are inside the flower (1)

(b) When a butterfly visits a type X flower to collect nectar, some pollen grains will stick on the middle portion of the mouthpart. (1)
When the same butterfly visits a type Y flower, (1)
the middle portion of the mouthpart rubs against the stigma which is at the middle position of the flower, achieving cross-pollination. (1)

(c) This avoid self-pollination / ensures cross-pollination. (1)
This result in more genetic variation among the offspring. (1)

(7 marks)

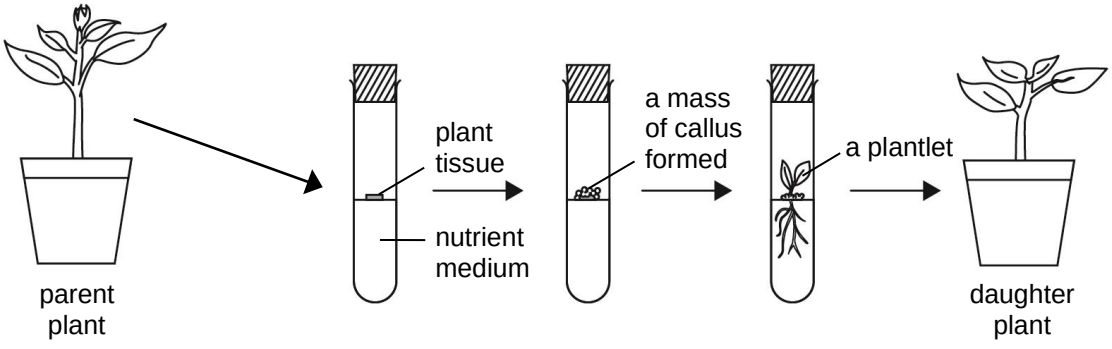
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	3
(c)	Applying concepts	2

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|!|BCELB0130000003|!| ★ **SI**

Micropropagation (a tissue culture technique) can be used to propagate plants. In this method, a small plant tissue was taken from the parent plant for growing new plants. The nutrient medium is sterilized before use and it provides essential materials for plant growth. The diagram below shows an outline of micropropagation.



(a) The nutrient medium contains sugars and mineral salts. Explain why sugars are added. (2 marks)

(b) What condition of a plant is required when choosing it as the parent plant for micropropagation? (1 mark)

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(c) Is the genetic make-up of the daughter plants the same as or different from that of the parent plant. Give a reason for your answer. (2 marks)

(d) Give ***two*** advantages of micropropagation over propagation using seeds. (2 marks)

##

(a) The plant tissue cannot carry out photosynthesis to make carbohydrates. (1)
It needs an external supply of sugars for the tissue cells to release energy and as raw materials for growth. (1)

(b) It should be healthy and free of diseases. (1)

(c) The genetic make-up of the daughter plants is the same as that of the parent plant (1)
because the daughter plants are formed by mitosis of the parent tissue cells. (1)

(d) This is a faster / a surer way of producing daughter plants. (1)
The desirable characteristics of the parent can be retained in the daughter plants. (1)
(Or other reasonable answers)

(7 marks)

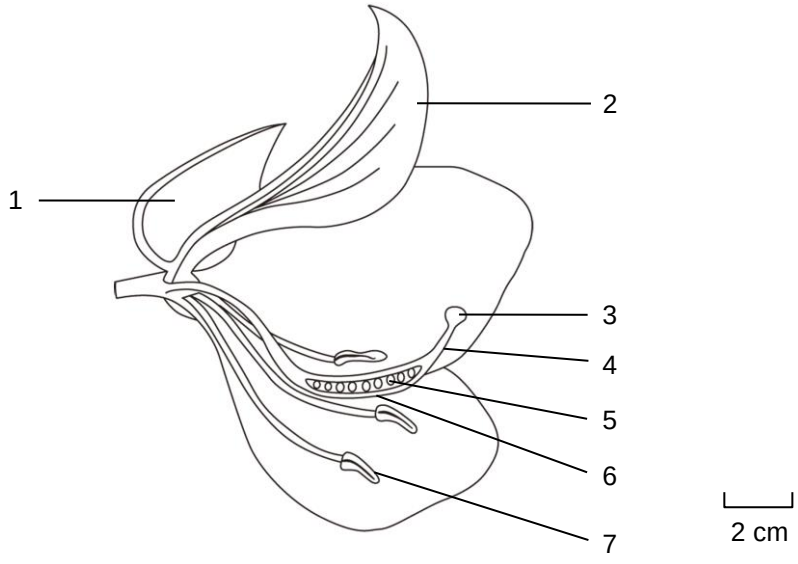
##

	Skill assessed	Mark(s)
(a)	Practical skills	2
(b)	Practical skills	1
(c)	Applying concepts	2
(d)	Applying concepts	2

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|!|BCELB0130000004|!| ★

The diagram below shows a longitudinal section of a flower.



(a) With reference to the diagram, state ***two*** structural features of the flower for insect pollination. (2 marks)

_____

_____

_____

_____

(b) Using the letters in the diagram, indicate the structure(s) in which

(i) meiotic cell division takes place. (1 mark)

_____

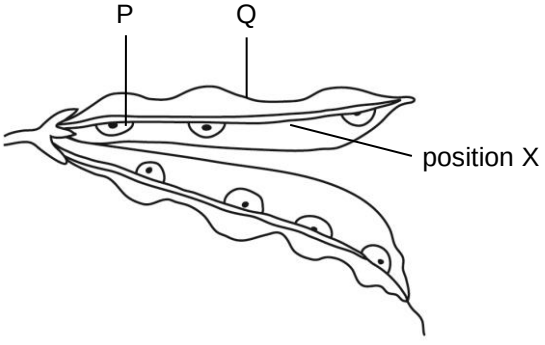
(ii) fertilization takes place. (1 mark)

_____

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(c) Describe the events leading to fertilization after a pollen grain landing on the stigma. (3 marks)

(d) The diagram below shows the fruit formed after fertilization.



(i) Name the parts of the flower from which P and Q have developed. (2 marks)

(ii) Explain why P is absent at position X. (1 mark)

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##

(a) The flower is large / petals are large. (1)

Anthers and stigma are enclosed inside the flower. (1)

(b) (i) 5, 7 (1)

(ii) 5 (1)

(c) When a pollen grain reaches a stigma, it is stimulated by the sugary solution on the stigma to develop a pollen tube. (1)

The pollen tube secretes enzymes to digest tissues of the style and the ovary, bringing the male gamete towards the ovule. (1)

The male gamete is released to fuse with the female gamete in the ovule. (1)

(d) (i) P: ovule (1)

Q: ovary wall (1)

(ii) The ovule at position X has not been fertilized. (1)

(10 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(c)	Understanding basic concepts	3
(d)(i)	Applying concepts	2
(d)(ii)	Applying concepts	1

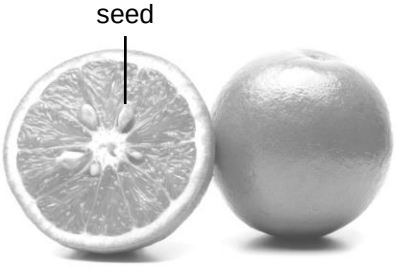
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|!|BCELB0130000005|!| ★

The photographs below show some onion bulbs and oranges. When an onion bulb and the seed of an orange were planted in moist soil, they developed into young plants.



onion bulbs



oranges

(a) Which type of cell division takes place during the early development of the young plants? (1 mark)

_____

(b) Explain why the young plant developed from a seed of an orange is genetically different from the parent plant. (3 marks)

_____

_____

_____

_____

_____

(c) If an onion bulb and a seed of an orange were planted in dry soil, the onion bulb would have a higher chance to develop new plants than the seed. Suggest why. (1 mark)

_____

_____

_____

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(d) Overcrowding of offspring is less likely to occur when planting a seed, but it is more likely to occur when planting an onion bulb. Explain why. (2 marks)

(e) If an onion bulb is evenly cut into pieces of the same shape and size, only some of the pieces can give rise to new plants when planted in favourable conditions. Suggest why. (1 mark)

##

(a) Mitotic cell division (1)

(b) Seeds are produced through sexual reproduction (1)
which involves the fusion of gametes. (1)
A seed contains a combination of genetic materials from the parents. (1)

(c) The onion bulb has a higher water content and the buds in the bulb can grow in the expense of the food reserve stored in the swollen scale leaves. (1)

(d) There are multiple buds inside an onion bulb. (1)
They may develop at the same time. (1)

(e) Only some cut pieces contain buds which can develop into new plants. (1)

(8 marks)

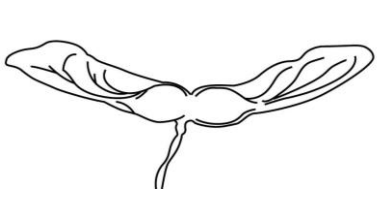
##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Applying concepts	3
(c)	Applying concepts	1
(d)	Applying concepts	2
(e)	Applying concepts	1

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|!|BCELB0130000006|!| ★

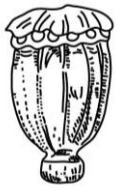
The diagram below shows three different fruits, P, Q and R.



P



Q



R

(a) State the difference(s) between a fruit and a seed with respect to

(i) their origins. (1 mark)

(ii) their functions. (2 marks)

(b) Fruits P and Q are dispersed by wind. Explain how they are adapted for wind dispersal. (4 marks)

(c) Fruit R is also dispersed by wind. There are small holes near the top of the fruit. When it is windy, the fruit shakes and the seeds inside it are thrown out. Explain how this is more advantageous than directly dropping the seeds under the parent plant. (3 marks)

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(d) Describe ***one*** way how animals may help with the dispersal of seeds and fruits, illustrated with an example. (3 marks)

##

(a) (i) A fruit is formed from the ovary, while a seed is formed from the ovule. (1)

(ii) Fruits protect the seeds and help disperse them. (1)

Seeds contain the embryos which can grow into new plants. (1)

(b) Fruit P has a wing-like structure (1)

which helps it float in the air. (1)

Fruit Q is small, light and have 'parachutes'. (1)

It can easily be blown away by wind. (1)

(c) When the fruit shakes, the seeds can be thrown fairly far away from the parent. (1)

This can reduce competitions between the offspring and parent. (1)

If the seeds are dropped directly under the parent plant, there will be great competition between the offspring and parent. (1)

(d) Any ***one set*** of the following: (3)

Some fruits, like burdock, (1)

have hooks or spines which cling to the fur of animals. (1)

When the animals move, the fruits are carried to other places. (1)

OR

Some fruits are fleshy and juicy to attract animals to eat them, such as grape and peach, (1)

The seeds of these fruits may have hard seed coat to withstand the digestive enzymes in the animals' intestines, so that they can be dispersed through the animals' droppings. (1)

Some animals may eat the fleshy parts of fruits and then discard the seeds to other places. (1)

(13 marks)

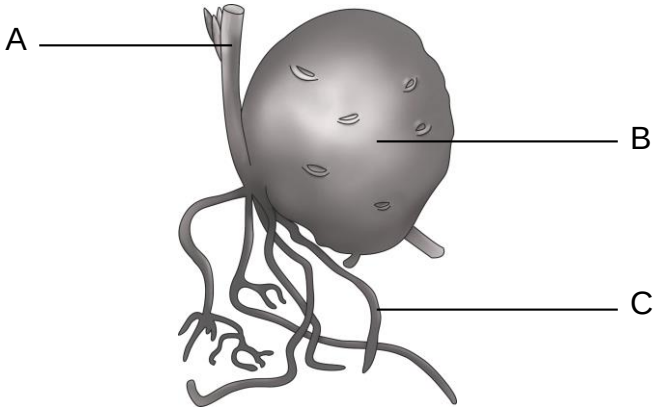
##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	2
(b)	Analysing data	4
(c)	Analysing data	3
(d)	Understanding basic concepts	3

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|!|BCELC0130000007|!|

Irene puts a potato tuber in a pot and covers it with wet soil. After several days, a short shoot grows out from the tuber.



(a) As the shoot grows up, structure B becomes smaller in size. Explain why. (3 marks)

(b) Name the tissue which transports

(i) food in structure A; (1 mark)

(ii) water in structure C. (1 mark)

(c) Describe how a new tuber is developed from this potato plant. (4 marks)

##

(a) The starch stored in structure B (1)
is used in respiration to release energy (1)
for the growth of the new plant. (1)

(b) (i) phloem (1)

(ii) xylem (1)

(c) Green leaves are developed and (1)

carry out photosynthesis to make food. (1)

The sugar is transported down by the phloem to the tip of an underground stem. (1)

Sugar is converted to starch which is the stored in the new tubers. (1)

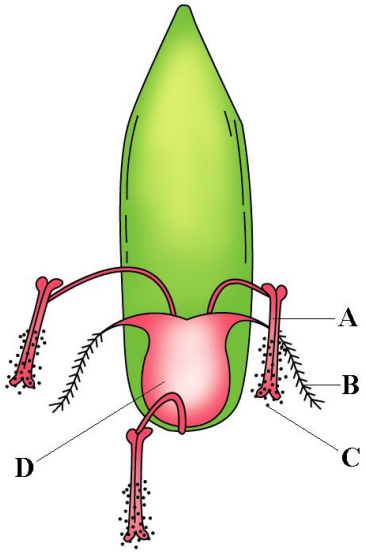
(9 marks)

##

	Skill assessed	Mark(s)
(a)	Applying concepts	3
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)	Applying concepts	4

|!|BCELC0130000008|!|

Tim observed some flowers in the school garden. He picked some of the flowers and dissected one flower in the laboratory. The following diagram shows what he observes under a dissection microscope.



(a) (i) Name structure A. (1 mark)

(ii) Explain how the position of structure A suggests that the flowers are wind-pollinated. (2 marks)

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(b) Name structure B. (1 mark)

(c) Explain why this flower produces a large amount of structure C, which is small and light. (2 marks)

(d) (i) Name structure D. (1 mark)

(ii) State the fate of structure D after fertilization. (1 mark)

##

(a) (i) anther (1)
(ii) The anthers hang outside the flower (1)
so as to release the pollen grains in the wind. (1)
(b) stigma (1)
(c) to compensate for the great number of pollen grains that cannot be transferred onto the stigma by wind (1)
to be easily carried away by wind (1)
(d) (i) ovary (1)
(ii) It develops into the fruit. (1)

(8 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	2
(b)	Analysing data	1
(c)	Applying concepts	2
(d)(i)	Analysing data	1
(d)(ii)	Understanding basic concepts	1

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|!|BCELC0130000009|!|

The following diagrams show a flower (**Diagram A**) and an underground structure (**Diagram B**) of a flowering plant.

(Diagrams A and B are not taken from the same plant species.)

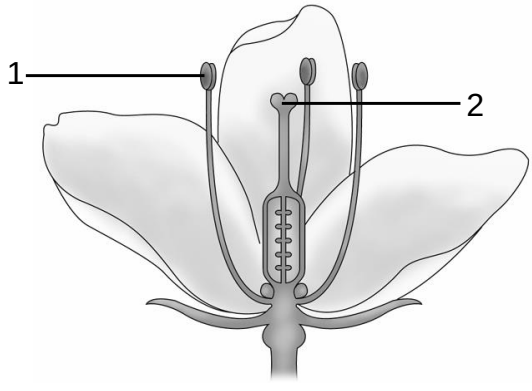


Diagram A

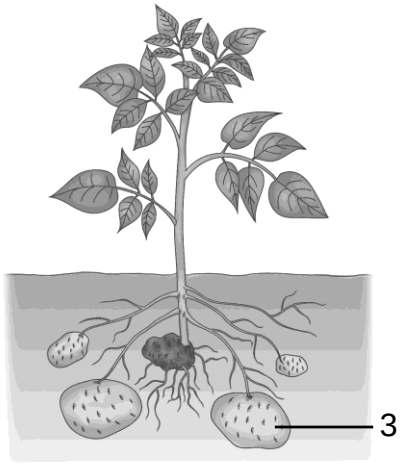


Diagram B

(a) (i) Name structure 1. (1 mark)

(ii) Describe how structures 1 and 2 are involved in the reproduction of the plant that involves self-pollination. (5 marks)

(b) (i) Name the type of asexual reproduction as shown in **Diagram B**. (1 mark)

(ii) Describe how structure 3 is formed. (3 marks)

(c) State ***two*** advantages of using the reproduction method as shown in **Diagram B** over that as shown in Diagram A. (2 marks)

##

(a) (i) anther (1)

(ii) Pollen grains are produced from structure 1 by meiotic cell division. (1)

Pollen grains are then transferred to structure 2. (1)

The sugary solution secreted by structure 2 stimulates the pollen grains to form pollen tubes. (1)

The pollen tube then enters the micropyle and releases the male gamete. (1)

The male gamete then fuses with the female gamete to form a zygote. (1)

(b) (i) vegetative propagation (1)

(ii) Green leaves carry out photosynthesis. (1)

Some food made in the leaves is passed down to the tip of an underground stem. (1)

The accumulation of food leads to growth of the new tuber (structure 3). (1)

(c) Any ***two*** of the following: (2)

- It allows an organism to reproduce quickly without the involvement of another parent. (1)
- Plenty of food reserve is available for the growth of the offspring. (1)
- It allows the species to survive adverse condition. (1)

(Or other reasonable answers)

(12 marks)

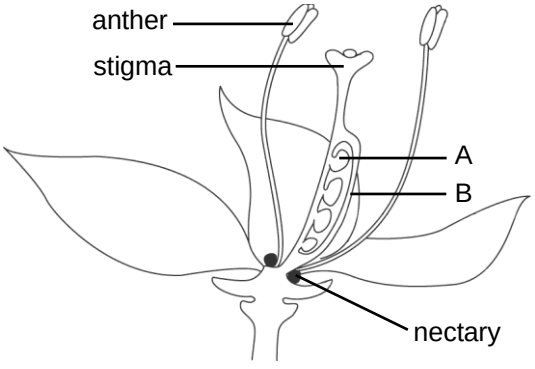
##

	Skill assessed	Mark(s)
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	5
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	3
(c)	Understanding basic concepts	2

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|!|BCELB0130000010|!| ★

The diagram below shows a flower of a certain plant species.



(a) Describe how to show that the plant only carries out cross-fertilization. State the procedures. (5 marks)

(b) Gametes are formed in the anther and structure A. State and explain if the following pairs are genetically identical.

(i) Gametes formed in the anther and gametes formed in structure A (3 marks)

(ii) Stigma and structure B (2 marks)

(c) Is this flower an insect-pollinated or wind-pollinated flower? Explain why.

(2 marks)

##

(a) Cut off the anthers in the flowers before they reach maturity. (1)

Brush pollen grains from the anthers of another plant of the same species onto the stigma of the flower (Flower X). (1)

Cover the plant with a plastic bag to prevent unwanted pollination. (1)

Repeat the steps by transferring pollen grains from the anthers to the stigma of a flower (Flower Y) in the same plant instead of using another plant. (1)

The formation of fruit from Flower X but not from Flower Y indicates that the plant only carries out cross-fertilization. (1)

(b) (i) They have different genetic make-up (1)

because they are formed by meiotic cell division (1)

which produces gametes with genetic variations through independent assortment / crossing over of homologous chromosomes. (1)

(ii) They have the same genetic make-up (1)

because they are derived from the same zygote. (1)

(c) Insect-pollinated flower (1)

This is because the flower contains nectaries that can attract insects for pollination. (1)

(12 marks)

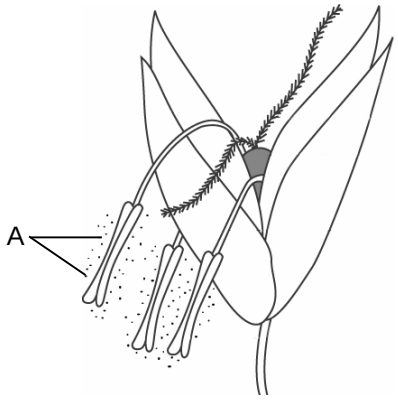
##

	Skill assessed	Marks
(a)	Applying concepts	5
(b)(i)	Applying concepts	3
(b)(ii)	Applying concepts	2
(c)	Analysing data	2

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|!|BCELC0130000011|!|

The diagram below shows the flower of a grass.



(a) With reference to the diagram, state ***two*** features that suggest that the flower above is wind-pollinated. (2 marks)

(b) (i) What will the ovary develop after fertilization? (1 mark)

(ii) State ***two*** roles of the structure developed from the ovary. (2 marks)

(c) (i) Which type of cell division is involved in the formation of structure A? (1 mark)

(ii) Describe the importance of this type of cell division to the plant. (3 marks)

##

(a) Any ***two*** of the following: (2)
- feathery stigma (1)
- stigma / anthers hang outside the petals (1)
- pollen grains are produced in large numbers (1)

(Or other reasonable answers)

(b) (i) The ovary will develop into a fruit after fertilization. (1)

(ii) The fruit helps the seed dispersal (1)
and protects the seeds. (1)

(c) (i) Meiotic cell division (1)

(ii) In meiotic cell division, haploid gametes are produced. (1)
The normal diploid chromosome number can be restored (1)
during fertilization / fusion of male and female gametes. (1)

(9 marks)

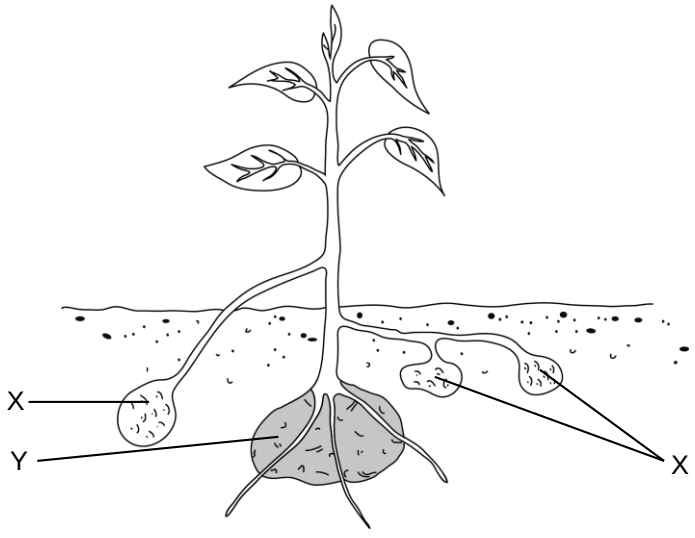
##

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	3

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|!|BCELB0130000012|!| ★

The diagram below shows a young potato plant.



(a) Which type of storage organ is structure Y? State the food that is stored in structure Y. (2 marks)

(b) (i) Describe and explain the changes in the size of structures X during the growth period of this plant. (2 marks)

(ii) Why does the size of structure Y change in a different way from that of structure X? (2 marks)

(c) State ***two*** disadvantages of reproduction by forming structures X when comparing with reproduction by forming flowers. (2 marks)

##

(a) Structure Y is a stem tuber. (1)
It stores starch. (1)

(b) (i) Structure X grows larger (1)
because the food manufactured by the leaves is passed down to structure X and stored. (1)

(ii) Structure Y becomes smaller. (1)
This is because the food stored in structure Y is used to develop the shoot and adventitious roots in the plant. (1)

(c) Any ***two*** of the following: (2)

- The offspring produced lack genetic variation and may not be able to adapt to environmental changes. (1)
- Overcrowding may occur. This leads to competition for space and resources among offspring and parents. (1)
- The chance of disease transmission from parent plants to offspring is higher. (1)

(Or other reasonable answers)

(8 marks)

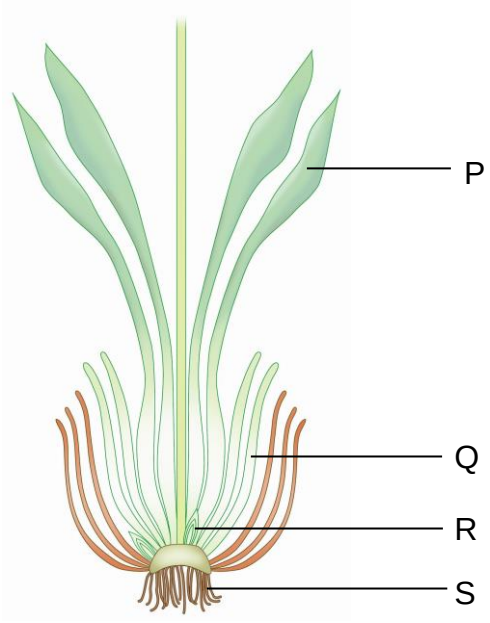
##

	Skill assessed	Marks
(a)	Analysing data	2
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2
(c)	Understanding basic concepts	2

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|!|BCELB0130000013|!| ★

The diagram below shows a growing onion bulb.



(a) State ***one*** difference between the cells in structures P and Q. (1 mark)

(b) Describe how structures P and S facilitate the growth of structure R. (3 marks)

(c) The onion can reproduce by forming new bulbs as well as by forming flowers.

(i) State ***two*** differences between the reproduction by onion bulb and reproduction by flower. (2 marks)

(ii) State ***one*** advantage for each reproduction method shown above. (2 marks)

##

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(a) The cells in structure P have chloroplasts while the cells in structure Q have not. (1)

(b) The food produced by structure P in photosynthesis is passed to structure R. (1)
Structure S absorbs water that is transported to structure R. (1)
Structure R obtains the food and water that can be used to support the mitotic cell division and growth of structure R. (1)

(c) (i) Any ***two*** of the following comparisons: (2)

Reproduction by onion bulb	***Reproduction by flower***
- involves one parent	- usually involves two parents
- does not involve gametes / fertilization	- involves gametes / fertilization
- involves mitotic cell division only	- involves both meiotic cell division and mitotic cell division
- produces genetically identical offspring	- produces genetically different offspring

(Or other reasonable answers)

(ii) Any ***one*** of the following advantages of reproduction by onion bulb: (1)

- It allows an organism to reproduce quickly without the involvement of another parent. (1)
- The storage organs contain plenty of food reserve for the growth of the offspring. (1)

(Or other reasonable answers)

Any ***one*** of the following advantages of reproduction by flowers: (1)

- Offspring show genetic variations which may help them adapt to environmental changes. (1)
- Seeds can be dispersed over long distances, reducing overcrowding and competition. (1)
- The chance of disease transmission from parent plants to offspring is lower. (1)

(Or other reasonable answers)

(8 marks)

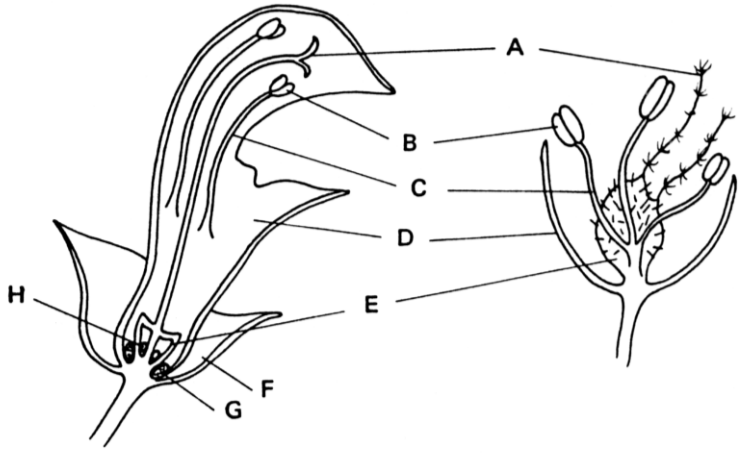
##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Understanding basic concepts	3
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	2

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The following diagram shows two kinds of flowers, X and Y. (They are not drawn to the same scale.)



(a) Which flower is an insect-pollinated flower? (1 mark)

(b) What would be the differences between these two flowers based on structures A, D and the products of structure B? (6 marks)

(c) (i) What is structure G? (1 mark)

(ii) State ***one*** function of structure G. (1 mark)

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(d) What will structures C, D, E and H change after fertilization? (3 marks)

##

(a) Flower X (1)

(b)

Structure	Flower X	Flower Y	
A (stigma)	- broad and sticky - enclosed in the corolla	- feathery - hang outside the flower	(2)
D (petal)	- larger petals	- smaller petals	(2)
Products of B (pollen grain)	- rough and sticky - fewer	- light and smooth - more	(2)

(c) (i) G is the nectary. (1)

(ii) It secretes nectar to attract insects. (1)

(d) After fertilization,

structures C and D will fall off; (1)

structure E will become the fruit wall; (1)

structure H will become a seed. (1)

(12 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Understanding basic concepts	6
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	1
(d)	Understanding basic concepts	3

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